Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 20 epochs with a batch size of 32. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the training loss.

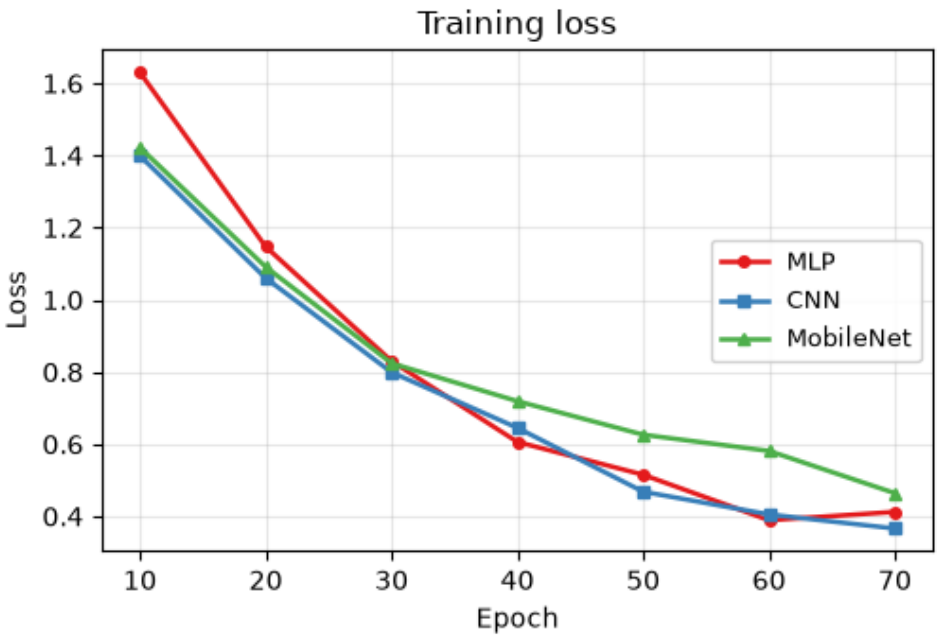


Figure 1: Training loss for the models in this experiment.

As shown in Figure 1, CNN has a loss of 0.644 at epoch 70. CNN has a loss of 1.260 at epoch 10. MobileNet has a loss of 0.765 at epoch 50 (Figure 1). At epoch 60, the loss of MLP is over 0.3 (Figure 1).

Figure 2 shows the accuracy by dataset.

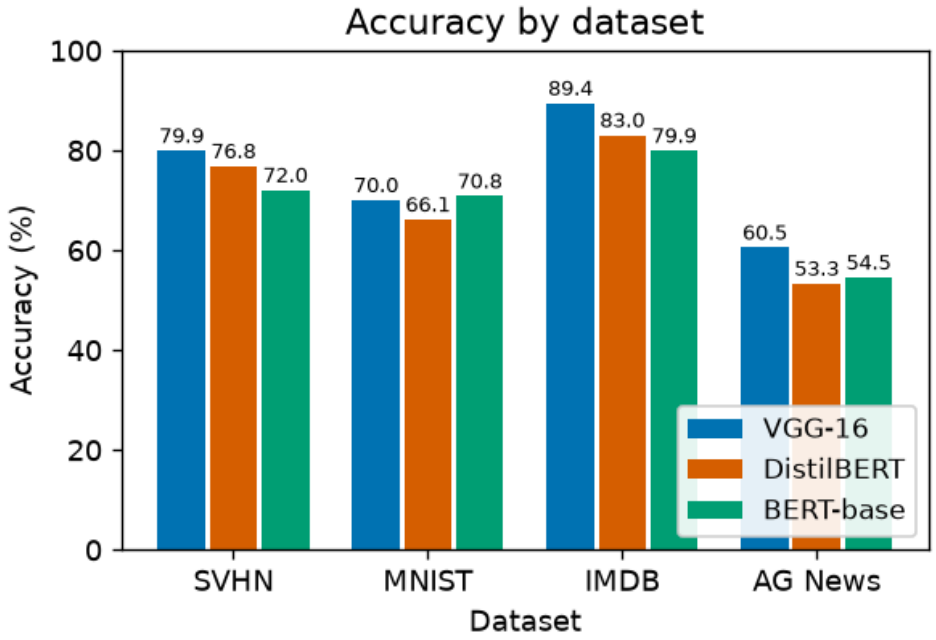


Figure 2: Accuracy by dataset for the models in this experiment.

VGG-16 reaches 72.0% accuracy on SVHN (Figure 2). DistilBERT reaches 53.3% accuracy on AG News. As shown in Figure 2, BERT-base achieves 71% accuracy on MNIST.

Figure 3 shows the f1 score during training.

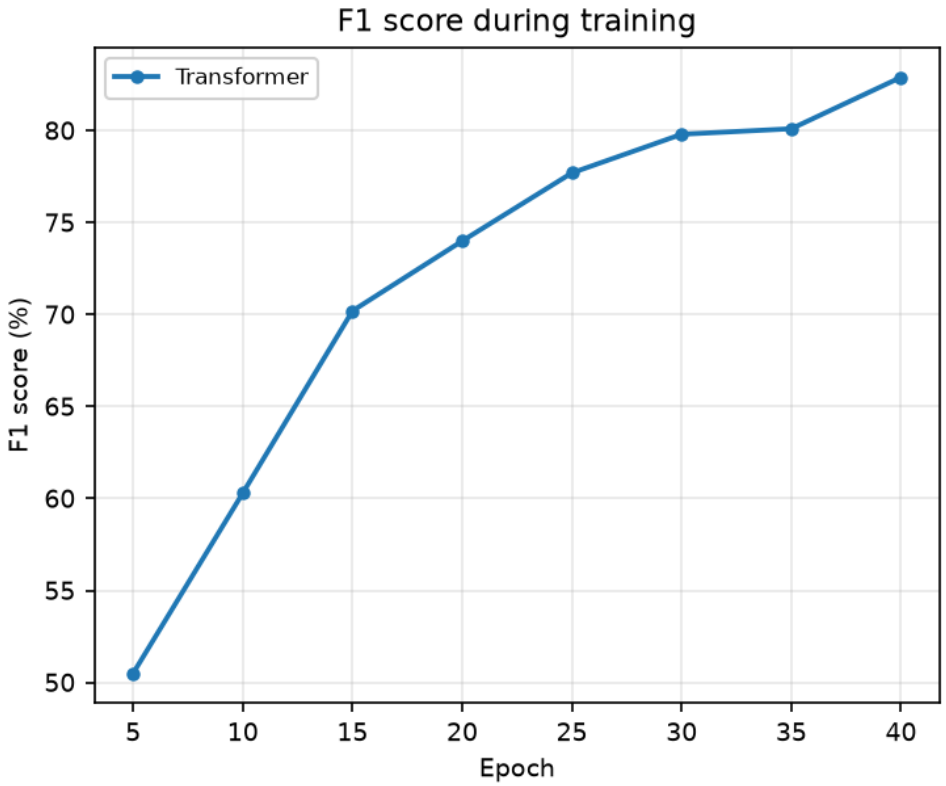


Figure 3: F1 score during training for the models in this experiment.

Transformer attains a F1 score of 80% at epoch 35. Transformer reaches about 80% F1 score at epoch 30. Transformer reaches 78.0% F1 score at epoch 30. Transformer scores 6.1 points higher on 35 than on 20.

Figure 4 shows the training loss.

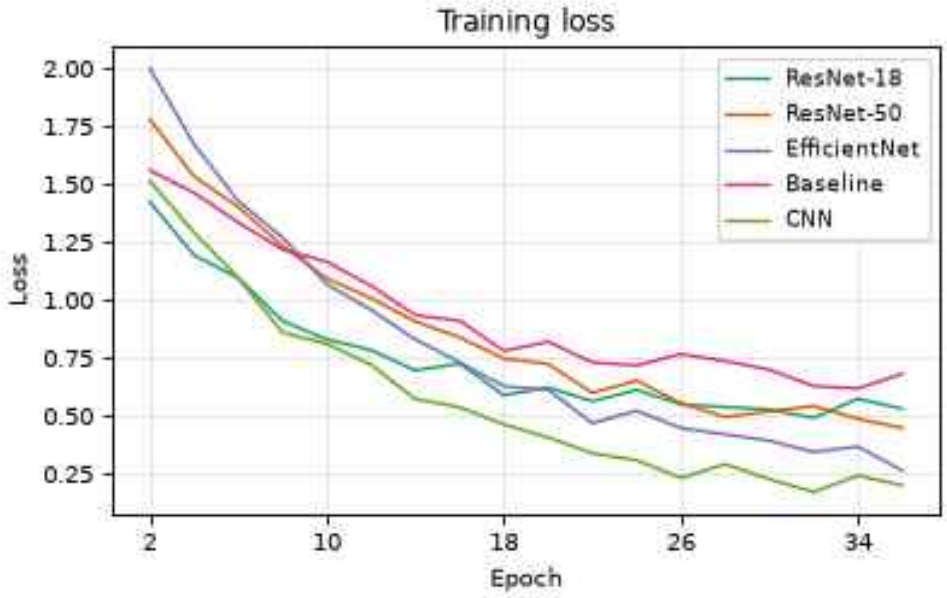


Figure 4: Training loss for the models in this experiment.

CNN has a loss of 0.57 at epoch 14. At epoch 34, the loss of EfficientNet is over 0.3 (Figure 4). As shown in Figure 4, ResNet-50 has a loss of 0.003 at epoch 12.

3 Discussion

On AG News, XGBoost obtains an accuracy of 70.9%. We note that the training was stable for all models.